3D Shape Classification with Multi-View Images REPORT

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Introduction

This report documents the architectural and implementation changes made to a multi-view 3D shape classification system. Two distinct deep learning paradigms, a **Convolutional Neural Network (CNN)** utilizing depthwise separable convolutions and a lightweight **Vision Transformer (ViT)**, were implemented and evaluated. The system is designed to classify furniture objects across **ten categories** using **twelve rendered views per shape**. This document details model design decisions, weight initialization strategies, training configurations, and the evaluation methodology employed to assess the trade-offs between these two architectures.

CNN ARCHITECTURE DESIGN

The CNN model was designed with an emphasis on parameter efficiency.

- ***Layer Structure:*** The architecture consists of an initial standard convolution block followed by two depthwise separable convolution blocks.
- ***Depthwise Separable Convolutions:*** These layers reduce the parameter count by performing per-channel convolutions followed by 1×1 pointwise convolutions.
- ***Classification Head:*** A global average pooling layer is used to convert feature maps into a fixed-size vector, which is then processed by a 1×1 convolution acting as a fully connected layer.

<u>RESULTS:</u>

```
============================================================
Per-Category Accuracy (mean pooling):
============================================================
bathtub        : 18/20 correct =  90.00%
bed            : 19/20 correct =  95.00%
chair          : 16/20 correct =  80.00%
desk           : 12/20 correct =  60.00%
dresser        : 19/20 correct =  95.00%
monitor        : 20/20 correct = 100.00%
night_stand    : 11/20 correct =  55.00%
sofa           : 15/20 correct =  75.00%
table          : 16/20 correct =  80.00%
toilet         : 18/20 correct =  90.00%
============================================================
Overall Test Error: 18.00% (Accuracy: 82.00%)

============================================================
Per-Category Accuracy (max pooling):
============================================================
bathtub        : 17/20 correct =  85.00%
bed            : 17/20 correct =  85.00%
chair          : 15/20 correct =  75.00%
desk           :  9/20 correct =  45.00%
dresser        : 19/20 correct =  95.00%
monitor        : 20/20 correct = 100.00%
night_stand    : 10/20 correct =  50.00%
sofa           : 11/20 correct =  55.00%
table          : 17/20 correct =  85.00%
toilet         : 17/20 correct =  85.00%
============================================================
Overall Test Error: 24.00% (Accuracy: 76.00%)
```

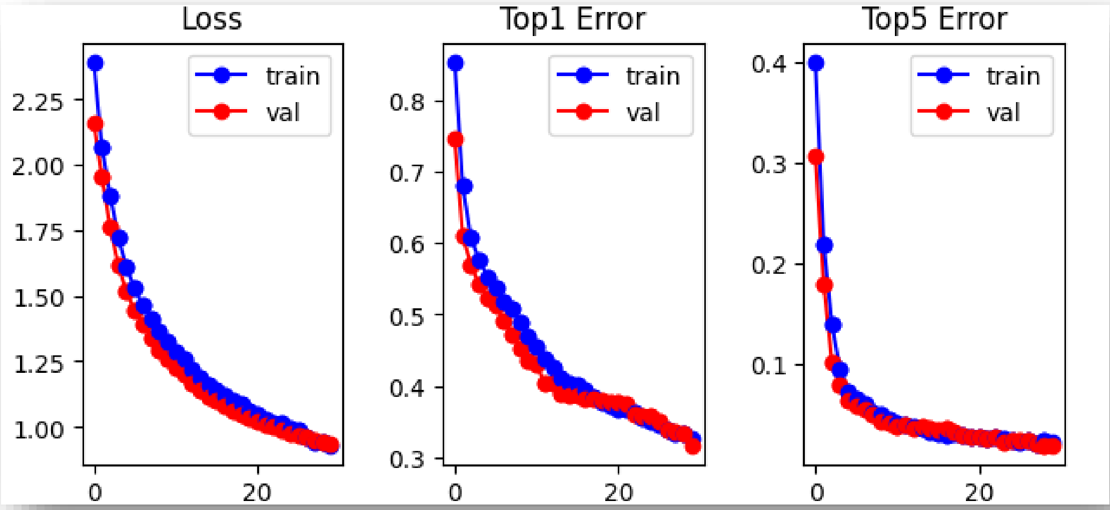

Figure 1: Loss & Error plots for CNN transformer

VISION TRANSFORMER ARCHITECTURE

The Vision Transformer (ViT) provides a global receptive field through its attention mechanism.

- ***Patch Embeddings:*** The model projects 112×112 images into an 8×8 grid of patches, resulting in 196 tokens.
- ***Transformer Encoder:*** A two-layer transformer encoder processes these tokens, utilizing multi-head attention and GELU activation.
- ***Embeddings:*** Learnable positional embeddings and a class token are included to provide spatial awareness and aggregate global features.

RESULTS:

```
============================================================
Per-Category Accuracy (mean pooling):
============================================================
bathtub        : 18/20 correct =  90.00%
bed            : 20/20 correct = 100.00%
chair          : 18/20 correct =  90.00%
desk           : 13/20 correct =  65.00%
dresser        : 16/20 correct =  80.00%
monitor        : 20/20 correct = 100.00%
night_stand    : 18/20 correct =  90.00%
sofa           : 16/20 correct =  80.00%
table          : 13/20 correct =  65.00%
toilet         : 20/20 correct = 100.00%
============================================================
Overall Test Error: 14.00% (Accuracy: 86.00%)

============================================================
Per-Category Accuracy (max pooling):
============================================================
bathtub        : 19/20 correct =  95.00%
bed            : 14/20 correct =  70.00%
chair          : 14/20 correct =  70.00%
desk           : 13/20 correct =  65.00%
dresser        : 16/20 correct =  80.00%
monitor        : 20/20 correct = 100.00%
```

```
night_stand     : 17/20 correct =  85.00%
sofa            : 14/20 correct =  70.00%
table           : 13/20 correct =  65.00%
toilet          : 19/20 correct =  95.00%
============================================================
Overall Test Error: 20.50% (Accuracy: 79.50%)
```

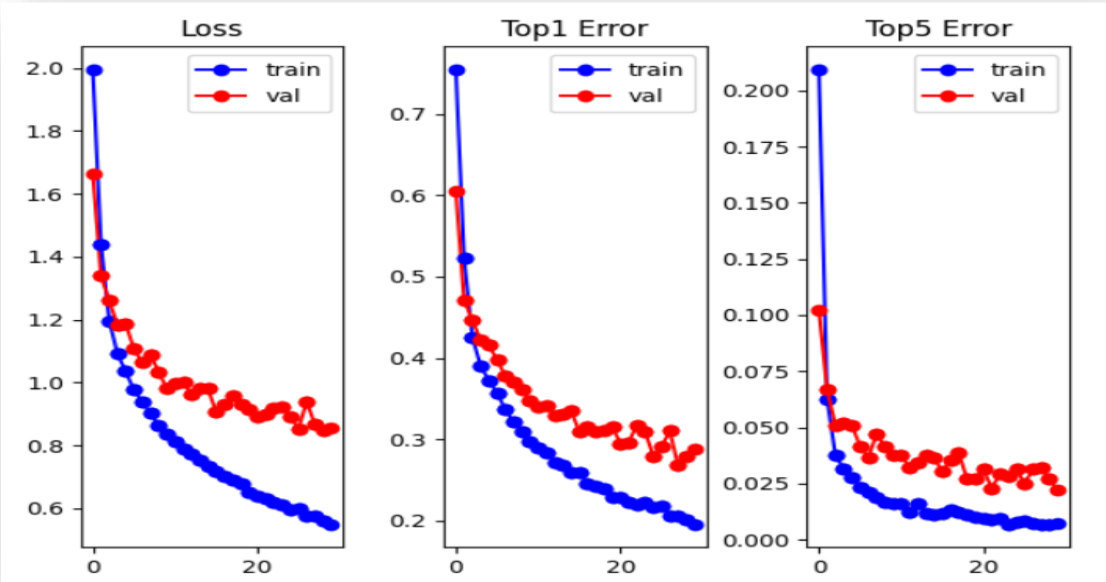

Figure 2: Loss & Error plots for vision transformer

Conclusion

This project illustrates the specific strengths of two modern vision architectures. While the CNN is better suited for smaller datasets due to its inherent inductive biases, the ViT demonstrates the adaptability of transformer models to multi-view classification tasks.